

PENTACHLOROPHENOL and 2,4-DINITROPHENOL

Synonyms

PCP (do not confuse with phencyclidine); DNP

Description

Includes:

pentachlorophenol and sodium salt of pentachlorophenate.

2,4-dinitrophenol (DNP) and related compounds (e.g. dinoseb).

Used as fungicides and wood preservatives. *Pentachlorophenol* is still currently registered as a wood preservative; use is restricted. *Dinitrophenol* was used medically for weight loss until its ban in 1930's due to toxicity; still promoted illegally as a fat burner and weight-loss agent and available via internet. Herbicide *dinoseb* is no longer registered for use in Canada; old supplies may still exist.

Toxicity

Mechanism of Toxicity

Pentachlorophenol and *dinitrophenol* are skin and mucous membrane irritants; may be corrosive at higher concentrations. Both agents uncouple oxidative phosphorylation causing an increase in metabolism, leading to hyperthermia. Toxicity resembles salicylate poisoning (see SALICYLATES). *Dinitrophenol* may cause methemoglobinemia and may have direct toxic effects on kidneys.

Toxic Dose

Not well established. Minimum oral lethal dose in humans estimated to be 29 mg/kg for *pentachlorophenol* and 20-50 mg/kg for *dinitrophenol*. Toxicity may be seen with 300 mg *dinitrophenol*.

Pentachlorophenol concentrations > 0.4% may be irritating to skin; concentrations > 10% may be corrosive. Dermal and inhalation exposure have resulted in fatalities. See Case Reports.

Case Reports

pentachlorophenol A 33-year-old began working in a chemical plant breaking large blocks of pentachlorophenol with a jackhammer. Within 1-2 weeks he developed lethargy and tachypnea which progressed over next 1-2 weeks to diaphoresis, Kussmaul-type respirations and thirst. On presentation to hospital he was alert with a RR of 44/minute, temperature of 37.8 degrees C, PO₂ 119, PCO₂ 17, pH 7.41. He became comatose with a RR of 56/minute and temperature of 40.5 degrees C. Patient was treated with IV fluids and ice packs but died approximately 1 hour later.

A 71-year-old ingested 120-240 mL of a weed killer containing 12% pentachlorophenol and 1.5% related compounds in a hydrocarbon base. Within 2 hours patient became restless and diaphoretic. He was lavaged for several hours until returns no longer had an odour. During this time, he became more restless, diaphoretic, agitated and hyperkinetic with some jerking movements. Temperature was 37.2 degrees C and RR was 20/minute. Phenytoin and dexamethasone were given for "cerebral excitation". On day 3 forced diuresis with furosemide and mannitol was begun and continued until day 5. Patient recovered uneventfully.

dinitrophenol A 17-year-old ingested 12-15 dinitrophenol capsules (32-40 mg/kg) sold as "diet pills". She vomited several times afterwards and became diaphoretic and jittery. She presented to hospital 4 hours post ingestion with temperature of 37.1 degrees C, HR 150/minute, BP 142/48 mmHg, and RR 42/minute. She was given activated charcoal and IV fluids. Temperature rose to > 40 degrees C over the next few hours despite physical cooling measures, fluids and antipyretics. She developed leg cramps and became agitated and combative, eventually becoming minimally responsive to pain. Patient developed ventricular tachycardia and asystole and died approximately 10 hours post ingestion.

A 22-year-old became agitated and withdrawn while taking dinitrophenol 600 mg/day for 4 days for weight loss. Sixteen hours after his last dose he appeared ill, diaphoretic, and feverish (38.9 degrees C). He became increasingly agitated and delirious despite midazolam and physical cooling. He became bradycardic and asystolic approximately 1 hour later and died.

A 30-year-old presented flushed, diaphoretic, tachycardic with yellow sclera and serum CK of 62,000. He admitted to taking dinitrophenol for weight loss and bodybuilding. Mental status deteriorated, temperature rose to 42.2 degrees C, and patient was intubated. Cooling blankets reduced temperature to 40 degrees C. Dantrolene was given and within 20 minutes temperature decreased to 38.2 degrees C; repeated doses of dantrolene were given over next 12 hours. Patient developed renal failure and remained intubated for 12 days but recovered without sequelae.

Pharmacokinetics

Onset of symptoms may be rapid or insidious. Both compounds are absorbed following oral, dermal and inhalation exposure.

pentachlorophenol Peak levels may occur within 4 hours after an oral dose. Mainly excreted unchanged in urine; small portions excreted in feces or metabolized to other compounds. Elimination half-life ranges from 13 hours to 20 days.

dinitrophenol Peak levels may occur 2-4 hours after oral exposure (animal data). Metabolized; parent compound and metabolites eliminated in urine. Elimination half-life estimated to be 5-14 days.

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Clinical Effects

- **Topical:** Irritation with > 0.4% solutions; burns with > 10% solutions. Dinitrophenol causes yellow staining of skin. Systemic effects may occur.
- **Ocular:** Irritation, burns.
- **Inhalation:** Dusts or mists may cause irritation to nose, throat and upper respiratory tract. Systemic effects may occur.
- **Ingestion:**
 - HEENT:** Irritation or burns to mouth and throat depending on concentration. Delayed development of cataracts have been described with use of dinitrophenol as a weight loss agent.
 - CVS:** Tachycardia, bradycardia, dysrhythmias, asystole, cardiovascular collapse.
 - Respiratory:** Tachypnea, respiratory failure. Pulmonary edema has been reported.
 - Neurologic:** Headache, dizziness, confusion, apprehension, agitation followed by seizures and coma.
 - GI:** Abdominal pain, nausea, vomiting or diarrhea may occur. May lead to volume depletion. Stools may be discoloured yellow by dinitrophenol.
 - Hepatic:** Possible hepatic toxicity.
 - GU:** Possible renal toxicity.
 - Fluids/Lytes/Acid-Base:** Anion gap metabolic acidosis (may be severe). Hyperglycemia or hypoglycemia may be seen.
 - Musculoskeletal:** Muscle weakness may occur; rhabdomyolysis has been reported.
 - Blood:** Leukocytosis may be seen with severe acute poisoning. Late-developing aplastic anemia has been associated with pentachlorophenol exposure (uncommon). Methemoglobinemia can occur experimentally with dinitrophenol; probably does not contribute significantly to toxicity.
 - Other:** Hyperthermia and diaphoresis are common (may lead to dehydration); chronic exposure is often associated with weight loss.

Treatment

1. **Topical:** Wash skin thoroughly with soap and water. Yellow staining of skin from dinitrophenol need not be removed entirely. Treat systemic symptoms as needed.
2. **Ocular:** Irrigate with copious amounts of tepid water or saline for 15 minutes. Obtain ophthalmological opinion if irritation persists.
3. **Inhalation:** Remove from exposure. Treat systemic symptoms as needed.
4. **Ingestion:** Administer activated charcoal in cases of recent ingestion (efficacy unproven).
5. Protect airway and assist ventilation as needed.
Caution: Intubation may be indicated in patients with deteriorating CNS status; however, intubation and ventilation can be associated with rapid worsening of toxicity and death unless hyperventilation is maintained.
6. Monitor vital signs, CBC, electrolytes, blood gases, glucose, ECG, renal and liver function tests, creatine kinase, urine for myoglobin.
7. Maintain fluid and electrolyte balance. Rehydrate with IV normal saline or D5½NS. Correct metabolic acidosis with IV sodium bicarbonate. Monitor potassium closely; correct hypokalemia as required. Avoid fluid overload due to risk of pulmonary edema.
8. Correct **hypoglycemia**. Low brain glucose can occur in the presence of normal blood glucose and may account for confusion, delirium, agitation. Maintain a high normal blood glucose level. Patients with neurologic symptoms should receive an IV bolus of 25-50 mL of 50% dextrose (repeat boluses may be required).
9. Treat **hyperthermia** aggressively with physical measures (e.g. cool mist, cooling blankets, ice packs, and fans). Do not use salicylates. Refractory hyperthermia may require dantrolene and/or benzodiazepine/barbiturate coma.
10. **Seizures** should be treated with bolus dose(s) of IV 50% dextrose and IV benzodiazepines. **Caution:** seizures may worsen metabolic acidosis and increase toxicity. Protect airway and hyperventilate. **Agitation, delirium or CNS depression** may respond to bolus dose(s) of IV 50% dextrose. Avoid sedatives unless required for severe agitation. **Caution** administering any medication which causes respiratory depression as suppression of hyperventilation may result in severe acidemia and death. Protect airway and hyperventilate.
11. Treatment of pulmonary edema is primarily supportive.
12. Exchange transfusion has been used successfully for pentachlorophenol poisoning in infants.
13. Hemodialysis and hemoperfusion do not appear effective. Hemodialysis may be required in patients with renal failure or severe acidosis refractory to conventional treatment.

KEY POINTS ON NEXT PAGE

Key Points

- ✓ Topical exposure can result in skin and mucous membrane irritation; corrosive injury with higher concentrations.
- ✓ Systemic toxicity resembles salicylate poisoning with hyperventilation, hyperthermia, diaphoresis, metabolic acidosis, agitation, seizures, and death.
- ✓ Treatment includes aggressive supportive care primarily directed at rehydration, temperature control with ice packs and possibly dantrolene, and control of seizures with IV dextrose and benzodiazepines.
- ✓ **Caution:** Intubation and ventilation can be associated with rapid worsening of toxicity and death *unless* hyperventilation is maintained.